Def. Let E be an Algebraic Extension of a field F.

Two elements $\alpha, \beta \in E$ are said to be Conjugate over F if: $m_{\alpha,F} = m_{\beta,F}$.
That is, they have same minimal monic irreducible polynomial.

Theorem. Let F be a field, and α, β be algebraic over F with $\deg(\alpha, F) = n$
α, β are Conjugate over F if and only if; a map

$$\psi_{\alpha,\beta} : F(\alpha) \to F(\beta) : \sum_{i=0}^{n-1} c_i \alpha^i \mapsto \sum_{i=0}^{n-1} c_i \beta^i \quad \text{where } c_i \in F$$

is Isomorphism

Proof. Suppose α, β are Conjugate over F with repect to $p(x) \in F[x]$.
By assumption, $\deg p(x) = n$. Now, the evaluation Homomorphisms

$$\phi_\alpha : F[x] \to F(\alpha) ; f(x) \mapsto f(\alpha),$$
$$\phi_\beta : F[x] \to F(\beta) : f(x) \mapsto f(\beta)$$

have the same Kernel $\langle p(x) \rangle$. And, there exist Natural Field Isomorphisms

$$\varphi_\alpha : F[x]/\langle p(x) \rangle \to F(\alpha) : f(x) + \langle p(x) \rangle \mapsto f(\alpha).$$
$$\varphi_\beta : F[x]/\langle p(x) \rangle \to F(\beta) : f(x) + \langle p(x) \rangle \mapsto f(\beta).$$

Define a map:

$$\psi_{\alpha,\beta} : F(\alpha) \to F(\beta) ; x \mapsto (\varphi_\beta \circ \varphi_\alpha^{-1})(x)$$

It is clearly Isomorphism, because $\psi_{\alpha,\beta} = \varphi_\beta \circ \varphi_\alpha^{-1}$, Composition of Isomorphism.
Since $F(\alpha)$ is a Vector space over F generated by $\{\alpha^i \mid 0 \le i \le n-1\}$,
$x \in F(\alpha)$ Can be represented as: for some $c_i \in F$,

$$x = \sum_{i=0}^{n-1} c_i \alpha^i$$

Now, $$\psi_{\alpha,\beta}(x) = (\varphi_\beta \circ \varphi_\alpha^{-1})\left(\sum_{i=0}^{n-1} c_i \alpha^i\right) = \sum_{i=0}^{n-1} c_i \cdot ((\varphi_\beta \circ \varphi_\alpha^{-1})(\alpha))^i = \sum_{i=0}^{n-1} c_i \cdot \beta^i.$$

Conversely, suppose the map $\psi_{\alpha,\beta}$ is Isomorphism,
let $m_{\alpha,F}(x) \in F[x]$ be an minimal monic irreducible polynomial of α. Then,

$$\psi_{\alpha,\beta}(m_{\alpha,F}(\alpha)) = m_{\alpha,F}(\beta) = 0.$$

That is, $m_{\beta,F}$ divides $m_{\alpha,F}$. Similarly $m_{\alpha,F}$ divides $m_{\beta,F}$.
Since both are monic, $m_{\alpha,F} = m_{\beta,F}$.

「Corollary. Let E/F be a fied, and $\alpha \in E$ be a algebraic over a field F.

If $\psi : F(\alpha) \to \overline{F}$ is 1-1 homomorphism such that $\forall a \in F$, $\psi(a) = a$,

then $\psi(\alpha)$, α are conjugate over F.

Conversely, if $\beta \in E$ is conjugate of α over F, then there exists unique 1-1 homomorphism

$$\psi_{\alpha,\beta} : F(\alpha) \to \overline{F}$$

which satisfies $\forall a \in F$, $\psi(a) = a$ and $\psi_{\alpha,\beta}(\alpha) = \beta$.

Def. Let E be a field, let $\sigma : E \to E$ be an Automorphism, $F \leq E$ be a subfield.
An element $a \in F$ is said to be *left fixed by σ* if: $\sigma(a) = a$.
A collection S of Automorphisms of E is said to be *leaves a subfield F of E fixed* if:
$$\forall \sigma \in S,\ \forall a \in F,\ \sigma(a) = a.$$

Thm. Let $\{\sigma_i \mid i \in I\}$ be a Collection of automorphisms of a field E.
Then, a set
$$E_{\{\sigma_i\}} \overset{\text{def}}{=} \{a \in E \mid \forall i \in I,\ \sigma_i(a) = a\}$$
forms a subfield of E. (In fact, $E_{\{\sigma_i\}}$ is called fixed field of $\{\sigma_i \mid i \in I\}$).

Proof. Let $a, b \in E_{\{\sigma_i\}}$ be given. Then, for any $i \in I$,
$$\sigma_i(a-b) = \sigma_i(a) - \sigma_i(b) = a-b.$$
$$\sigma_i(ab) = \sigma_i(a)\sigma_i(b) = ab$$
$$\sigma_i(a^{-1}) = \sigma_i(a)^{-1} = a^{-1}$$
Clearly $0, 1 \in E_{\{\sigma_i\}}$.

$$\begin{array}{c} E \\ | \\ E_{G(E/F)} \\ | \\ F \end{array}$$

$$\begin{array}{c} Aut(E) \\ | \\ G(E/F) = \{\sigma \in Aut(E) \mid \forall a \in F,\ \sigma(a) = a\} \end{array}$$

7, 20, 22 c, 34, 36

48. 7.

observe that: $\sqrt{1+\sqrt{2}} = x \Rightarrow \underbrace{x^4-2x^2-1}_{p(x)} = 0$

It is irreducible in $\mathbb{Q}$, because: If $p(x)$ has a factor of degree 1, $p(\pm 1)=0$. This is false.

Thus, if $p(x)$ is reducible, then it has a decomposition as:

$$x^4-2x^2-1 = (x^2+ax+b)(x^2+cx+d) \qquad a,b,c,d \in \mathbb{Q}.$$

But, this is impossible because the system

$$\begin{cases} a+c=0 \\ ac+b+d=-2 \\ ad+bc=0 \\ bd=-1 \end{cases}$$

has no rational solution. Now, we know that $\operatorname{irr}(\sqrt{1+\sqrt{2}}, \mathbb{Q}) = p(x)$.

And, $p(x)$ can be split over $\mathbb{C}$ as:

$$(x^4-2x^2-1) = (x^2-(1+\sqrt{2}))(x^2-(1-\sqrt{2}))$$

$$= (x-\sqrt{1+\sqrt{2}})(x+\sqrt{1+\sqrt{2}})(x-\sqrt{1-\sqrt{2}})(x+\sqrt{1-\sqrt{2}})$$

Now, $-\sqrt{1+\sqrt{2}}, \pm\sqrt{1-\sqrt{2}}$ are conjugate with respect to $\sqrt{1+\sqrt{2}}$.

48.20. Since τ_5, τ_3, τ_2 are Automorphisms, so is $\tau_5 \circ \tau_3 \circ \tau_2$.

τ_2, τ_3, τ_5 fix $\mathbb{Q}$, thus $E_{\tau_5 \circ \tau_3 \circ \tau_2}$ contains $\mathbb{Q}$.

But, $E_{\tau_5 \circ \tau_3 \circ \tau_2}$ dont contains $\sqrt{2}, \sqrt{3}, \sqrt{5}$, because

$$(\tau_5 \circ \tau_3 \circ \tau_2)(\sqrt{2}) = (\tau_5 \circ \tau_3)(-\sqrt{2}) = -\sqrt{2}$$

$$(\tau_5 \circ \tau_3 \circ \tau_2)(\sqrt{3}) = (\tau_5 \circ \tau_3)(\sqrt{3}) = \tau_5(-\sqrt{3}) = -\sqrt{3}$$

$$(\tau_5 \circ \tau_3 \circ \tau_2)(\sqrt{5}) = -\sqrt{5}$$

Since $\mathbb{Q}$ is the largest subfield of $\mathbb{Q}(\sqrt{2},\sqrt{3},\sqrt{5})$ not containing $\sqrt{2}, \sqrt{3}, \sqrt{5}$, thus $E_{\tau_5 \circ \tau_3 \circ \tau_2} = \mathbb{Q}$.

#48.22. c)

Clearly, $\tau_2^2 = \tau_3^2 = \tau_5^2 = 1$ (identity map)

#48.36.

Note: for any prime p,

$$\Phi_p(x) = \frac{x^p-1}{x-1} = \sum_{i=0}^{p-1} x^i$$

is irreducible over $\mathbb{Q}$, let α be a zero of Φ_p.

a). α^i $(1 \le i \le p-1)$ are distinct zeros of Φ_p.

proof. For any $1 \le j \le p-1$

$$\Phi_p(\alpha^j) = \sum_{i=0}^{p-1} \alpha^{j \cdot i} = \left(\sum_{i=0}^{p-1} \alpha^i\right) \cdot \alpha^{j \cdot i - i} = 0.$$

And, if for some $1 \le j < k \le p-1$, $\alpha^j = \alpha^k$, then $\alpha^{k-j} = 1$. That is, $x^{k-j}-1$ divides Φ_p, it contradicts to Φ_p is irreducible.

Since $\deg \Phi_p = p-1$, these are all the zeros of Φ_p.

b) $G(\mathbb{Q}(\alpha)/\mathbb{Q})$ is Abelian of order $p-1$.

By a), $\{\alpha^i \mid 1 \le i \le p-1\}$ are conjugate over $\mathbb{Q}$.

Let $\varphi \in G(\mathbb{Q}(\alpha)/\mathbb{Q})$ be given. Then, φ leaves $\mathbb{Q}$ fixed, and $\varphi(\alpha) = \alpha^i$ for some $1 \le i \le p-1$ since the corollary. Moreover, this φ should be uniquely determined by choice of $1 \le i \le p-1$,

order of $G(\mathbb{Q}(\alpha)/\mathbb{Q})$ is $p-1$.

And, let $\varphi_i \in G(\mathbb{Q}(\alpha)/\mathbb{Q})$ be an Automorphism s.t $\varphi_i(\alpha) = \alpha^i$.

Then, for any $1 \le i, j \le p-1$,

$$\varphi_i \circ \varphi_j(\alpha) = \alpha^{i+j} = \alpha^{j+i} = \varphi_j \circ \varphi_i(\alpha)$$

Thus $G(\mathbb{Q}(\alpha)/\mathbb{Q})$ is Abelian.

c). Since $\deg \Phi_p = p-1$, $[\mathbb{Q}(\alpha) : \mathbb{Q}] = p-1$

And, $\{\alpha^i \mid 1 \le i \le p-1\}$ is linearly independent, because for $c_i \in \mathbb{Q}$,

$$\sum_{i=1}^{p-1} c_i \alpha^i = 0 \Rightarrow c_1 = \cdots = c_{p-1} = 0$$

(If not all zero c_i satisfies $\sum_{i=1}^{p-1} c_i \alpha^i = 0$, then by Cancelation, there exists a $f(x) \in \mathbb{Q}[x]$ s.t $\deg f < p-1$ and $f(\alpha) = 0$, which is contradiction.) Now, $\mathbb{Q}(\alpha) = \langle \{\alpha^i \mid 1 \le i \le p-1\} \rangle_{\mathbb{Q}}$.

Now, since each α^i $(1 \le i \le p-1)$ is left fixed by $\varphi \in G(\mathbb{Q}(\alpha)/\mathbb{Q})$ leaves $\mathbb{Q}$.

「Isomorphism Extension Theorem.

Let E be an Algebraic Extension of a field F, and $\sigma: F \to F'$ be an Isomorphism.
Then, there exists an extension of σ:

$$\tau : E \to \overline{F'}$$

which is 1-1 and Homomorphism, and satisfies $\forall a \in F$, $\tau(a) = \sigma(a)$.

Proof.

「Thm. Let E be a finite extension of a field F.
Let σ be an Isomorphism of F onto F'.

$$\sigma(a_{i0}) + \sigma(a_{i1})x + \cdots + \sigma(a_{im_i})x^{m_i}$$

$$\tau(a_{i0}) + \tau(a_{i1})\tau(\alpha_i) + \cdots + \tau(a_{im_i})\tau(\alpha_i)^{m_i}$$

$$= \tau(a_{i0} + a_{i1}\alpha_i + \cdots + \quad) = \tau(0) = 0,$$

#49.2.

By theorem, there exist $\{E : \mathbb{Q}(\sqrt{2},\sqrt{15})\} = [E : \mathbb{Q}(\sqrt{2},\sqrt{15})] = 2$ extensions.

Denote $F = \mathbb{Q}(\sqrt{2},\sqrt{15})$. since $\mathrm{irr}(\sqrt{3},F) = x^2-3$, if τ is an extension of σ leaving F fixed,

$$\tau(\sqrt{3}) = \pm\sqrt{3}.$$

if $\tau(\sqrt{3})=\sqrt{3}$, then

$$\tau(\sqrt{2})=\sqrt{2},\quad \tau(\sqrt{3})=\sqrt{3},\quad \tau(\sqrt{5}) = \tau(\sqrt{15}\cdot(\sqrt{3})^{-1}) = \sqrt{15}\cdot(\sqrt{3})^{-1} = -\sqrt{5}$$

if $\tau(\sqrt{3})=-\sqrt{3}$, then

$$\tau(\sqrt{2})=\sqrt{2},\quad \tau(\sqrt{3})=-\sqrt{3},\quad \tau(\sqrt{5})=\sqrt{5}.$$

#49.3.

Similarly, there exist $[E : \mathbb{Q}(\sqrt{30})] = 4$ extensions. Let $F = \mathbb{Q}(\sqrt{30})$.

Since $E = F(\sqrt{2},\sqrt{3}) = \{a_0 + a_1\sqrt{2} + a_2\sqrt{3} + a_3\sqrt{6} \mid a_0,a_1,a_2,a_3 \in F\}$

($\sqrt{2}\cdot\sqrt{3}\cdot\sqrt{30} = 6\sqrt{5} \Rightarrow \sqrt{5} \in F(\sqrt{2},\sqrt{3})$, thus $E = F(\sqrt{2},\sqrt{3})$)

and $\mathrm{irr}(\sqrt{2},F) = x^2-2$, $\mathrm{irr}(\sqrt{3},F) = x^2-3$, the extension τ is determined by

$\tau(\sqrt{3})$ \ $\tau(\sqrt{2})$	$\sqrt{2}$	$-\sqrt{2}$
$\sqrt{3}$	$-\sqrt{5}$	$\sqrt{5}$
$-\sqrt{3}$	$\sqrt{5}$	$-\sqrt{5}$

(The values in this table indicate $\tau(\sqrt{5})$.)

#49.5. Since $\mathrm{irr}(\sqrt[3]{2},\mathbb{Q}) = x^3-2$, $\mathrm{irr}(\sqrt{3},\mathbb{Q}) = x^2-3$,

there exist 6 extensions, which is determined by roots of x^3-2 and x^2-3.

#49.6.